

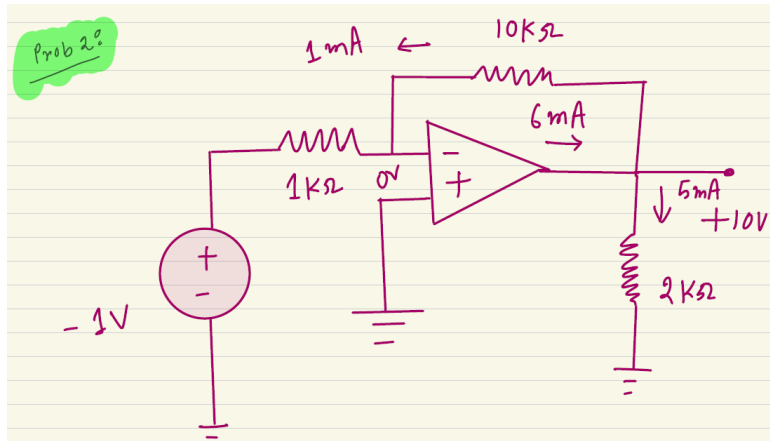
HW 5:

Prob 1:

$$V_o / V_{in} = -5$$

$$R_{in} = 20K\Omega$$

Prob 2:



Prob 3:

$$I_1 = 0.1mA$$

$$I_2 = 0.1mA$$

$$I_3 = 0.01A$$

$$R_L = 1200\Omega$$

$$V_L = -11.v \text{ from } -2.01v$$

Prob 4:

$$R_2 = 100K\Omega$$

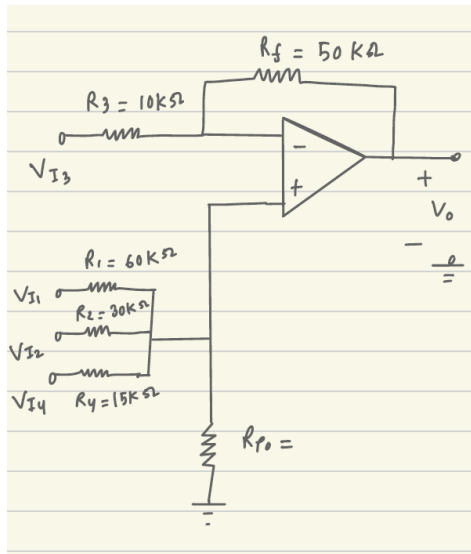
$$R_3 = 1.02K\Omega$$

$$\frac{V_o}{V_{in}} = -2.48v/v$$

Prob 5:

$$R = 100K\Omega$$

Prob 6:

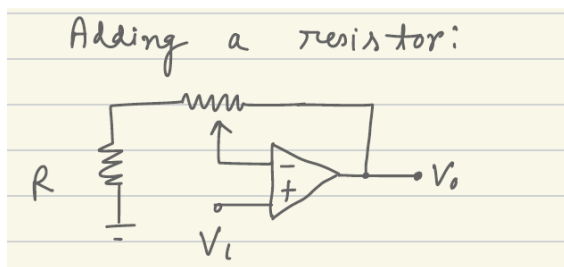


$$R_{P0} = 7.43\text{ K}$$

Prob 7:

$$\frac{V_o}{V_{in}} = \left(\frac{R_2}{R_1} + 1 \right) \left(\frac{R_4}{R_3 + R_4} \right)$$

Prob 8:



$$R = 1\text{ K}\Omega$$

Prob 9:

The range: -10V to +10V

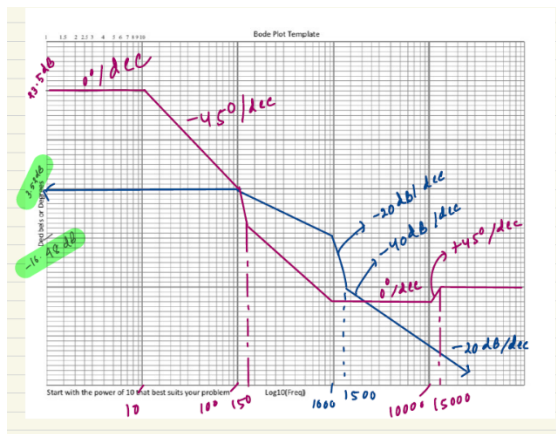
Prob 11:

a) $\frac{V_0}{V_{in}} = 2$

b) $\frac{V_0}{V_{in}} = -6$

Prob 12:

$$\frac{V_{out}}{V_{in}} = \frac{3 + 2j\omega R_f C_f}{2(1 + j\omega R_1 C_1)(1 + j\omega R_f C_f)}$$



Prob 15:

$$\frac{V_{out}}{V_{in}} = -4.75V / V$$

$$R_{in} = 10.2K\Omega$$

Prob 16:

$$V_{out} = -15V$$

$$V_{out} = 25017.19V$$